

END TERM EXAMINATION

THIRD SEMESTER [B.TECH] DECEMBER 2024

Paper Code: ECC-207

Subject: Digital Logic & Computer Design

Time: 3 Hours

Maximum Marks: 60

Note: Attempt five questions in all including Q. no. 1 which is compulsory. Select one question from each unit.

Q. No.	Question	Max. Marks	CO(s)
1. (a)	Convert $(1111000100)_2$ to Gray code, XS-3, and BCD.	4	CO1
(b)	Implement a half adder using NAND gates only.	4	CO2
(c)	What is the race-around condition in flip-flops? How can it be overcome?	4	CO3
(d)	Differentiate between a Ring counter and a Johnson counter.	3	CO3
(e)	What is an Input-Output Processor (IOP), and how does it contribute to the efficiency of I/O operations?	3	CO4
(f)	Differentiate between RISC and CISC architectures.	2	CO4
UNIT – I			
2. (a)	Simplify the following Boolean function using the Quine–McCluskey method: $f(A, B, C, D) = \sum m(1, 2, 4, 5, 7, 8, 10, 11, 12, 14)$	6	CO1
(b)	Find a minimal SOP expression using K-Map and draw the logic circuit of the minimal expression: $f(A, B, C, D) = \sum m(1, 3, 4, 5, 9, 10, 11) + d(6, 8)$	4	CO1
3. (a)	Design a 4-bit magnitude comparator circuit. Provide the truth table and logic design for the comparator.	5	CO2
(b)	Design a 16-to-1 multiplexer using 8-to-1 multiplexers. Provide the logic diagram and truth table for your design.	5	CO2
UNIT – II			
4. (a)	Explain the concept of a modulus counter. Design a synchronous counter with a modulus of 9 using JK flip-flops.	5	CO3
(b)	How does a Serial-In-Parallel-Out (SIPO) shift register differ from a Parallel-In-Serial-Out (PISO) shift register? Consider a 4-bit serial-in, serial-out (SISO) shift register with an initial state of 1101. Assume a clock signal triggers the shift operation on each rising edge. The input data stream is 1010. Illustrate the state transitions of the shift register for each clock cycle, showing the output after each shift operation. After four clock cycles, what will be the final state of the shift register?	5	CO3

[P.T.O.]

Q. No.	Question	Max. Marks	CO(s)
5. (a)	Design a sequence detector which detects 1100 (non-overlapping) from any given sequence. Implement using D flip-flops.	5	CO3
(b)	Describe the architecture of a Programmable Array Logic (PAL) device. How does it differ from a PLA in terms of internal structure and functionality?	5	CO2
UNIT – III			
6. (a)	Explain the role of the Control Unit, ALU, and registers in the CPU. How do they coordinate together to execute instructions?	5	CO4
(b)	Describe arithmetic micro-operations in the context of computer organization. Provide examples of arithmetic operations and explain how they are performed at the micro-operation level.	5	CO4
7. (a)	Define assembly language and describe its relationship with machine language.	5	CO4
(b)	Explain the concept of microprogrammed control in computer architecture. How does it differ from hardwired control, and what are its major advantages?	5	CO4
UNIT – IV			
8. (a)	Discuss the principles of cache mapping, including direct-mapped, set-associative, and fully associative cache mapping techniques.	5	CO4
(b)	Describe the role of Direct Memory Access (DMA) in input-output operations.	5	CO4
9. (a)	Define virtual memory. How does virtual memory address the physical memory limitations in modern computer systems?	5	CO4
(b)	Discuss the binary subtraction process and describe how borrow is handled in binary subtraction.	5	CO1

**** End of Question Paper ****